



RPKI

Securing BGP Route Announcements

Cryptographic proof that an AS is authorized to announce a prefix.

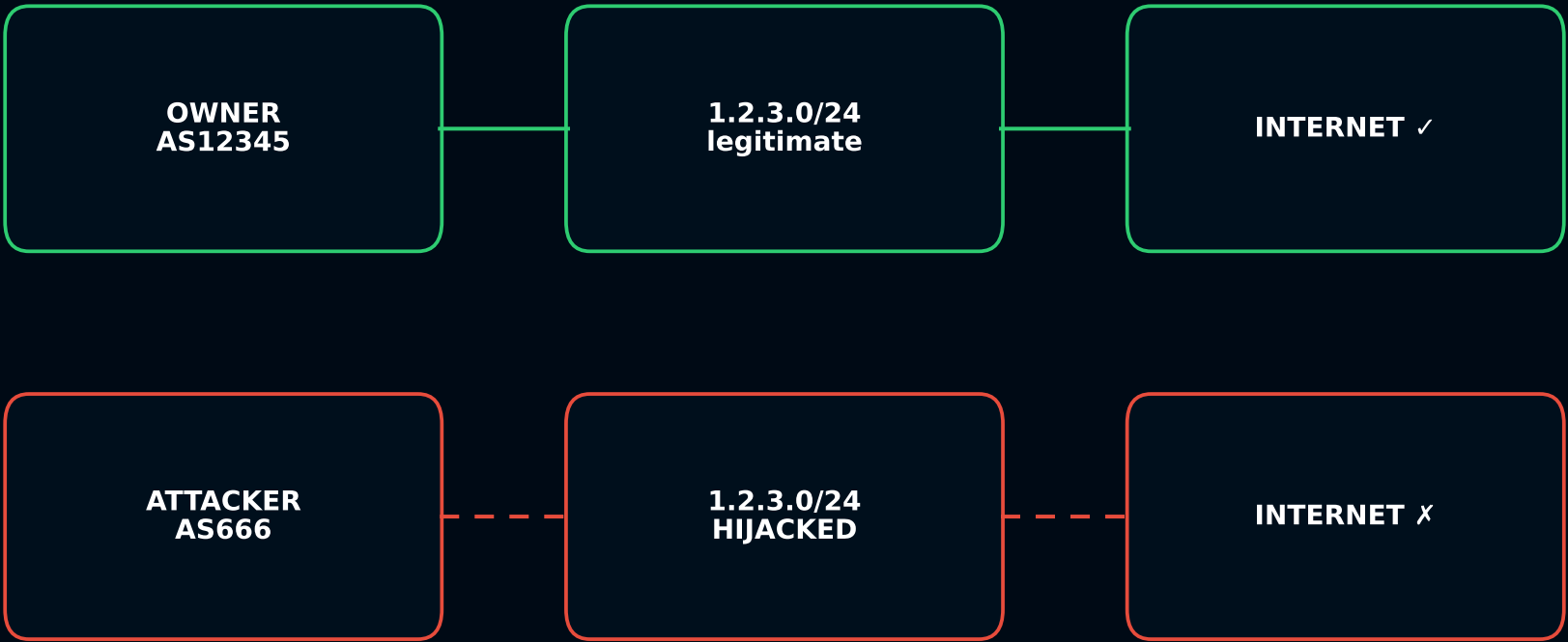
github.com/titan101/job_aids

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Created by Varun Sinha

THE PROBLEM

BGP was designed on trust. Any AS can announce any prefix.

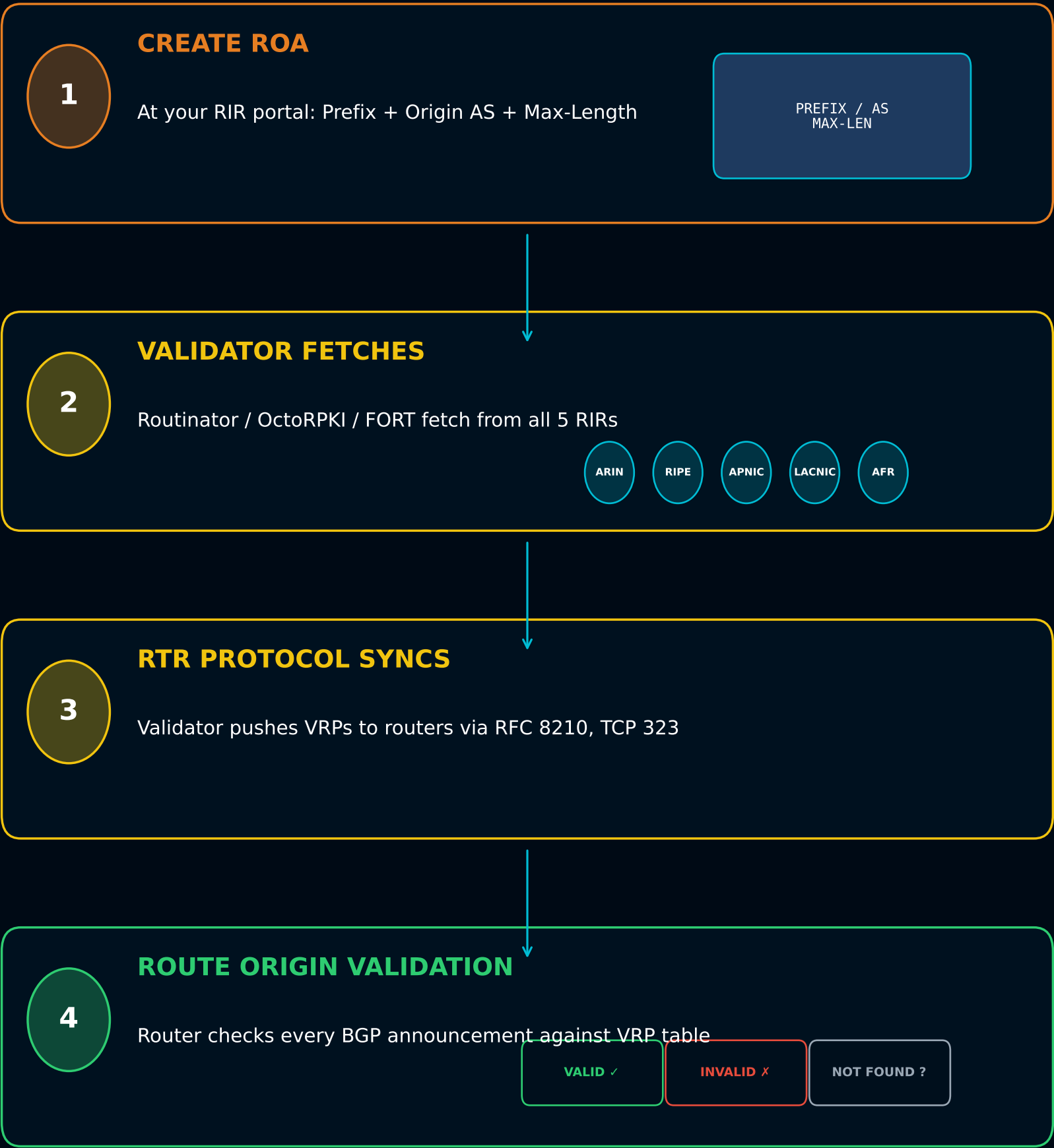


Without RPKI - both can be accepted. Traffic can go anywhere.



HOW RPKI WORKS

From ROA creation to route validation - 4 steps



THE 3 RPKI STATES

✓ **VALID**

Origin AS matches ROA AND prefix length $\leq$ max-length

-> **ACCEPT** and **PREFER** in route selection



✗ **INVALID**

Wrong origin AS OR prefix more specific than max-length

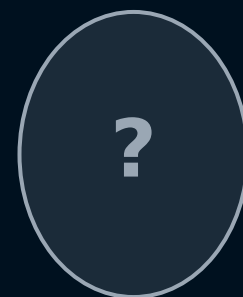
-> **DROP** - cryptographically proven wrong



? **NOT FOUND**

No ROA exists for this prefix - no assertion either way

-> **ACCEPT** and tag for monitoring



DEPLOYING RPKI

8 steps from zero to enforcing

1

Audit all prefixes you currently announce to BGP

inventory first

2

Log in to your RIR portal - enable Hosted RPKI

free at every RIR

3

Create ROAs - max-length = actual announced length

avoid blackholes

4

Deploy RPKI validator on your infrastructure

run at least two

5

Connect routers to validator via RTR protocol

TCP 323 / 3323

6

Enable ROV in MONITOR-ONLY mode

log, do not drop

7

Fix any of your own prefixes showing INVALID

clean your house

8

Switch to enforcement - DROP INVALID routes

phase carefully

Misconfigured ROAs can black-hole your own traffic.

KNOW THE LIMITS

X

AS_PATH attacks
That's BGPsec
not widely deployed

X

Route leaks
Legitimate AS,
wrong upstream

X

Path plausibility
ASPA objects
still emerging

X

Full path signing
BGPsec RFC 8205
near-zero production

VERIFICATION COMMANDS

Syntax varies by platform

- 1. Show RTR/RPKI session status
- 2. Show validated ROA cache
- 3. Show BGP route RPKI state
- 4. Debug validation events
- 5. Clear/refresh RTR cache
- 6. Verify specific prefix state
- 7. Show origin validation table



Save this.

RPKI protects BGP from origin hijacking.

57% of IPv4 prefixes have ROAs. 70% of traffic validated.

Still no RPKI? You're trusting every peer on the internet.

57%

IPv4 prefixes
with ROAs

70%

traffic
VALID (2025)

FREE

Hosted RPKI
at every RIR

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RPKI Reference | BGP Route Origin Validation | Platform-Agnostic